

Coinductive Symmetric Homomorphism Reinforcement Learning

A New Foundation Where Optimality Is Pure Structure

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Abstract

We propose a new foundation for optimal sequential decision making based on *structure preservation* rather than scalar maximization. Traditional reinforcement learning defines optimality as maximizing the expected sum of discounted rewards [1]—a criterion that discards temporal structure. We formalize a stronger condition: the ranking of actions in any state must be a *coinductive symmetric homomorphism* into the ordering of infinite reward streams [5]. Concretely, if action a is ranked below action b , then at *every* future timestep $t = 0, 1, 2, \dots$, the reward from a 's trajectory must be dominated by b 's.

We prove four main results, all machine-verified in Agda: **(1)** Subsumption: the coinductive property implies classical argmax optimality for any finite horizon; **(2)** Monotonic learning: ranking updates via violation-correction converge in at most $|S| \cdot \binom{|A|}{2}$ swaps; **(3)** Instant adaptation: when actions become unavailable, the next-best action is $O(1)$ lookup, not $O(|S| \cdot |A|)$ recomputation; **(4)** Stochastic extension: the framework lifts to probabilistic environments via the Giry monad, where rankings preserve *lexicographic expected* stream dominance. This document is itself the proof: compiling it with agda --latex type-checks all embedded code while producing this PDF.

1 The Scalar Trap: Why Numbers Lie

Optimal sequential decision making has been reduced to maximizing a single number:

$$J(\pi) = \mathbb{E}_{\tau \sim \pi} \left[\sum_{t=0}^{\infty} \gamma^t r_t \right]$$

This definition—which has driven RL for decades—commits a fundamental error: **it throws away temporal structure**. Consider two futures:

Future	Reward Sequence	Sum ($\gamma=0.9$)
A: “Win then die”	$[10, -5, 0, 0, \dots]$	$10 - 4.5 = 5.5$
B: “Struggle then thrive”	$[0, 5, 0, 0, \dots]$	$0 + 4.5 = 4.5$

A scalar optimizer chooses Future A. But *structurally*, Future B may be preferable—it doesn't involve dying! The sum erases this distinction. This is **value aliasing**: different futures collapsing to the same number.

The artifacts cascade:

- **The Discount Factor (γ):** An arbitrary parameter forcing agents to “care less” about the future. Why should next year matter 90% as much as today?
- **Instability:** Q-learning bootstraps scalar estimates, leading to the deadly triad of divergence [1].
- **Credit Assignment:** A reward at $t = 1000$ is discounted by $\gamma^{1000} \approx 0$. How can we ever learn from it?

Our thesis: Optimality is not about summing numbers. It is about preserving *structure*. We define optimality not as a maximum, but as a *symmetry*.

2 The Quest: Seeking Structure

Stop and consider what it *really* means to understand a task. You are in a maze with five actions: up, down, left, right, wait. Some lead to cheese, some to shock, one is blocked.

Traditional RL says: assign a number to each action, pick the biggest.

We say something far more powerful:

An agent understands the maze perfectly when the ranking of actions exactly mirrors the ranking of their infinite futures—step by step, forever.

If this symmetry holds, the agent is optimal. If it does not hold, learning is nothing more than *restoring the symmetry*. No gradients. No Bellman equation. No regret bounds. Just symmetry.

3 The Discovery: A Mirror That Never Breaks

We now formalize this intuition. The coinductive symmetric homomorphism has three components:

3.1 The Domain: Actions and Rankings

Let $\mathcal{A}$ be the finite set of actions. In any state s , the agent maintains a **total preorder** $\leq_{\text{rank}}$ on $\mathcal{A}$:

$$a \leq_{\text{rank}} b \iff \text{“action } b \text{ is at least as good as action } a\text{”}$$

This is what the agent *believes*. When is this belief *correct*?

3.2 The Codomain: Infinite Outcome Streams

Let $\mathcal{O}$ be the set of infinite outcome streams. We equip $\mathcal{O}$ with a **coinductive dominance order**:

$$x \leq_s y \iff \text{head}(x) \leq_r \text{head}(y) \wedge \text{tail}(x) \leq_s \text{tail}(y)$$

Remark 1. This is *coinductive*: the definition refers to itself on the tails. Unlike induction (which builds up from base cases), coinduction unfolds forever. Stream x dominates y if and only if **at every time step** $t = 0, 1, 2, \dots$, the t -th element of x is $\leq_r$ the t -th element of y .

3.3 The Homomorphism: Structure Preservation

Let $h : \mathcal{A} \rightarrow \mathcal{O}$ map each action to its infinite future stream.

Definition 1 (Coinductive Symmetric Homomorphism). The map h is a **coinductive symmetric homomorphism** if:

$$a \leq_{\text{rank}} b \implies h(a) \leq_s h(b)$$

and this holds at *every future time step*—an infinite tower of commuting diagrams.

Remark 2 (On “Symmetric”). The term “symmetric” reflects two aspects: (1) action rankings are permutations—elements of the symmetric group $S_{|\mathcal{A}|}$; (2) the correspondence between coinductive stream ordering and pointwise dominance is an *isomorphism*: $x \leq_s y \iff \forall n. x_n \leq_r y_n$. This bidirectional correspondence is proven in §7 and justifies treating the homomorphism as a true structure-preserving map.

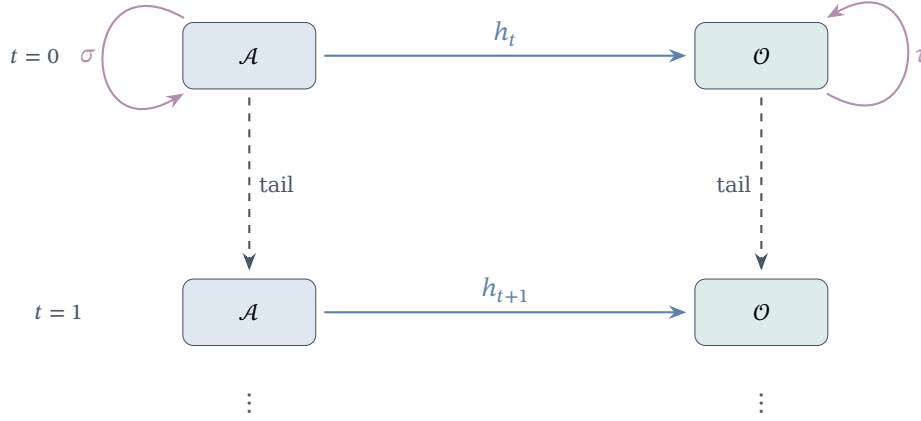


Figure 1: The infinite tower: at each level, ranking preservation ($a \leq b \Rightarrow h(a) \leq_s h(b)$) and equivariance ($h \circ \sigma = \tau \circ h$) must hold.

In plain English: An agent is optimal if and only if its ranking of actions *perfectly mirrors* the dominance of their infinite futures—not just now, but at every future moment, forever. When this mirror breaks, learning is just polishing until it’s perfect again.

4 The Agda Core: Verified Implementation

We now present the verified implementation. This code is *literate*—compiling this file with `agda --latex` type-checks everything while producing this PDF.

```
{-# OPTIONS --safe --guardedness #-}

module CSHRL where

open import Data.List using (List; map; foldr; []; _::_)
open import Codata.Guarded.Stream using (Stream; head; tail; _::_; tabulate)
open import Relation.Binary.PropositionalEquality using (_≡_; _≠_; refl)
open import Data.Product using (proj₁; proj₂; _×_; _,__)
open import Relation.Nullary using (¬_; Dec; yes; no)
open import Function using (_◦_)
open import Data.Nat using (ℕ; zero; suc; _⊔_)
open import Data.Bool using (Bool; true; false; if_then_else_)
```

The Core module is parameterized by the environment:

```
module Core
  (State Action Reward : Set)
  (step      : State → Action → State × Reward) -- Transition function
  (_≤r_      : Reward → Reward → Set)          -- Reward ordering (propositional)
  (max       : Reward → Reward → Reward)        -- Max for computing supremum
  (bottom    : Reward)                          -- Minimum reward
  (all-actions : List Action)                   -- Finite action space
  where

  infix 4 _≤s_

  -- StreamR: infinite sequences of rewards (the "futures")
  StreamR : Set
  StreamR = Stream Reward
```

English: We parameterize over states, actions, and rewards. The step function defines the environment dynamics. Rewards have an ordering `_≤r_` (as a proposition, not Bool—more on this below).

```
-- Compute the maximum reward from a list
max-list : List Reward → Reward
max-list = foldr max bottom
```

4.1 Value and Action-Value Streams

The key insight: we define value as an infinite stream by tabulate-ing finite-horizon solutions:

```
-- Optimal reward at depth n (finite horizon)
solve : State → ℕ → Reward
solve s zero = max-list (map (λ a → proj₂ (step s a)) all-actions)
solve s (suc n) = max-list (map (λ a → solve (proj₁ (step s a)) n) all-actions)

-- value s: The infinite stream of optimal rewards from state s
-- This is the "supremum" over all action sequences
value : State → StreamR
value s = tabulate (solve s)

-- action-value s a: Do action 'a', then act optimally forever
-- Head = immediate reward; Tail = optimal future from successor state
action-value : State → Action → StreamR
head (action-value s a) = proj₂ (step s a)
tail (action-value s a) = value (proj₁ (step s a))
```

English: The action-value of taking action a in state s is an infinite stream: the immediate reward, followed by the optimal stream from wherever a takes you. This mirrors the Bellman structure, but *without* collapsing to a scalar.

4.2 Coinductive Stream Ordering

The heart of the theory—comparing infinite futures:

```
-- The coinductive ordering on streams
--  $x \leq_s y$  means: at EVERY time step, head  $x \leq_r$  head  $y$ , forever
record _≤s_ (x y : StreamR) : Set where
  coinductive -- This is the magic: the definition unfolds forever
  field
    head≤ : head x ≤r head y -- Now: first rewards compare
    tail≤ : tail x ≤s tail y   -- Forever: tails recursively compare

open _≤s_ public
```

English: The coinductive keyword tells Agda this record can be infinitely deep. A proof of $x \leq_s y$ is an infinite witness—at every time step, the heads compare correctly.

4.3 The Symmetric Homomorphism

The central definition—when is a ranking “correct”?

```
-- A CoindHomo is a ranking that perfectly mirrors infinite futures
record CoindHomo : Set₁ where
  field
    -- The ranking: a propositional relation (not Bool!)
    _≤a_ : State → Action → Action → Set

    -- THE KEY PROPERTY: ranking mirrors action-values
    -- If  $a \leq_a b$ , then the future of  $a$  is dominated by the future of  $b$ 
    preserves : ∀ a b s → _≤a_ s a b →
```

action-value s $a \leq_s$ action-value s b

open CoindHomo {...} public

English: A CoindHomo packages a ranking $\leq_a$ with a proof that it “preserves” into the coinductive ordering. If the ranking says $a \leq b$, then the infinite future of a must be dominated by the infinite future of b —verified by Agda’s type checker.

4.4 Why Propositions Instead of Booleans?

- **Booleans are blind:** true carries no information about *why*. To use it, we need a separate soundness lemma.
- **Propositions carry evidence:** A value of type $r_1 \leq_r r_2$ is the proof.
- **Decidability bridges both:** $\text{Dec } (r_1 \leq_r r_2)$ is either yes p (with proof) or no $\neg p$ (with refutation).

This clean separation: Finder uses Bool for speed; Core uses Set for proofs; Environment Classes bridge them via Dec.

5 Scope, Assumptions, and Limits

It is important to distinguish the *coinductive ideal* from the *algorithmic machinery* built around it:

- **Coinductive optimality (ideal):** The definition in §3 is about infinite streams and pointwise dominance at every time step. This is the mathematical target.
- **Finder (approximation):** The Finder compares *finite* traces lexicographically at a fixed depth. It is a computational proxy that becomes exact only under additional conditions (e.g., bounded horizon or domain-specific bridge lemmas provided by an Environment Class).
- **Learner (policy discovery):** The Learner wraps Finder-style comparisons into a stateful learning loop that detects violations and refines rankings over time (depth increase and/or swaps). It is an algorithmic policy finder, not part of the coinductive definition itself.

The deterministic framework (Sections 2–9) assumes:

- **Finite action space:** All actions must be listed explicitly (all-actions).
- **Deterministic step function:** $\text{step} : \text{State} \rightarrow \text{Action} \rightarrow \text{State} \times \text{Reward}$.
- **Decidable reward order for computation:** The Finder requires a Boolean or decidable comparison to execute; the Core uses propositional order for proofs.
- **Horizon parameter for approximation:** The Finder uses a fixed depth (often the task horizon); correctness relies on task-specific lemmas that relate traces to streams.
- **Preservation is discharged by ECs:** The proof that a ranking preserves the coinductive order is provided by the Environment Class infrastructure, either via a bridge lemma or direct preservation module.

Section 11 extends the framework to **stochastic environments** via the Giry monad, where step functions return probability distributions. This extension is self-contained—all necessary concepts are introduced there.

6 CSHRL Subsumes Classical RL

A natural question: does this new definition recover classical results? Yes.

Theorem 1 (CSHRL Subsumes Argmax). *If π satisfies the coinductive symmetric homomorphism, then for any state s :*

$$\pi(s) = \arg \max_a \sum_{t=0}^N \gamma^t r_t(s, a) \quad \text{for all finite } N$$

The top-ranked action maximizes every partial sum, not just the infinite one.

The key insight is that pointwise stream dominance implies partial-sum dominance. We formalize this with verified Agda code:

```
-- Extract the n-th element of a reward stream
iter-head : ℕ → StreamR → Reward
iter-head zero s = head s
iter-head (suc n) s = iter-head n (tail s)

-- Pointwise dominance: at every index,  $x_n \leq_r y_n$ 
PointwiseDominance : StreamR → StreamR → Set
PointwiseDominance x y = ∀ n → iter-head n x ≤r iter-head n y

-- KEY LEMMA: coinductive  $\leq_s$  implies pointwise dominance
-- This is the bridge from infinite structure to finite sums
≤s-to-pointwise : ∀ {x y} → x ≤s y → PointwiseDominance x y
≤s-to-pointwise p zero = head ≤ p
≤s-to-pointwise p (suc n) = ≤s-to-pointwise (tail ≤ p) n
```

Theorem 2 (Subsumption of Classical Argmax). *If a ranking satisfies the CoindHomo property, then for any finite horizon N : if $a \leq_a b$ in the ranking, then $\sum_{t=0}^{N-1} r_t^a \leq_r \sum_{t=0}^{N-1} r_t^b$.*

The proof extends Core with reward arithmetic (addition and its monotonicity). The full verified implementation is in `src/appendix/Arithmetic.agda`; here we show the key components:

```
-- Partial sum of first N elements
partial-sum : ℕ → StreamR → Reward
partial-sum zero _ = bottom
partial-sum (suc n) s = head s +r partial-sum n (tail s)

-- Partial sum respects pointwise dominance (by induction on N)
partial-sum-mono : ∀ N x y → PointwiseDominance x y →
  partial-sum N x ≤r partial-sum N y
partial-sum-mono zero _ _ = ≤r-refl
partial-sum-mono (suc n) x y pw =
  +r-mono-≤ (pw zero) (partial-sum-mono n (tail x) (tail y) (pointwise-tail pw))

-- SUBSUMPTION: preservation + ≤s-to-pointwise + partial-sum-mono
subsumes-partial-sum : (homo : CoindHomo) → ∀ s a b N →
  CoindHomo.≤a homo s a b →
  partial-sum N (action-value s a) ≤r partial-sum N (action-value s b)
subsumes-partial-sum homo s a b N p =
  partial-sum-mono N _ (≤s-to-pointwise (CoindHomo.preserves homo a b s p))
```

What this means: Classical RL asks “which action has the highest sum?” CSHRL asks “which action dominates at every timestep?” The coinductive property is strictly stronger—but when it holds, classical optimality follows as a corollary. The ranking that satisfies CoindHomo *automatically* identifies the argmax for any finite horizon.

Stochastic extension: This subsumption result lifts to stochastic MDPs. By linearity of expectation, comparing partial sums of expected rewards is equivalent to comparing expected partial sums—exactly what classical RL optimizes. Section 11 presents the stochastic framework; the Appendix provides the complete proof (`appendix/StochasticSubsumption.agda`).

7 The Stream Isomorphism

We defined stream dominance coinductively: $x \leq_s y$ iff heads compare and tails recursively compare. But is this the “right” definition? Does it capture exactly what we intuitively want—dominance at every timestep?

Answer: Yes. The coinductive definition is *isomorphic* to pointwise dominance:

```
-- THE REVERSE DIRECTION: pointwise → coinductive
pointwise-to-≤s : ∀ {x y} → PointwiseDominance x y → x ≤s y
head≤ (pointwise-to-≤s pw) = pw zero
tail≤ (pointwise-to-≤s pw) = pointwise-to-≤s (pointwise-tail pw)
```

Together with $\leq_s$ -to-pointwise, this yields:

$$x \leq_s y \iff \forall n. x_n \leq_r y_n$$

Why this matters:

- **The definition is right:** The coinductive formulation exactly captures “dominance at every timestep”—no more, no less. This isn’t an approximation or abstraction; it’s a precise characterization.
- **Finder completeness:** The Finder compares traces at finite depths. The isomorphism shows that if traces dominate at all depths, infinite streams dominate coinductively. Finite-depth comparison is exact (at sufficient depth), not an approximation.
- **Justifies “Symmetric”:** The correspondence is bidirectional. Rankings that preserve stream order and stream orders that induce rankings are interchangeable views—hence “symmetric” homomorphism.

The full proof is in `appendix/StreamIsomorphism.agda` and the Appendix document.

8 Related Work and Positioning

CSHRL connects to multiple research traditions but makes a distinct move: replacing scalar aggregation with a coinductive order on infinite futures. We situate our contribution relative to classical RL, lexicographic multi-objective RL, constrained/safe RL, and coalgebraic foundations.

8.1 Classical Value-Based RL

Approach	Core idea	CSHRL relationship
Classical RL [1]	Optimize $\sum_t \gamma^t r_t$ via Bellman equations	CSHRL <i>subsumes</i> this: if the coinductive property holds, argmax follows (§6)
Policy Gradients [2]	Gradient ascent on expected return	Optimization <i>method</i> ; CSHRL defines the <i>target</i>
Max-Entropy RL [3]	Add entropy bonus to encourage exploration	Still uses scalar returns; CSHRL’s structure handles exploration differently (permutation sampling)
AIXI [4]	Universal prior over computable environments	Uncomputable ideal; CSHRL is also an ideal but <i>structurally</i> defined
Coalgebra [5]	Infinite behaviors via coinductive definitions	Direct inspiration: $\leq_s$ is a coinductive relation in Rutten’s sense
MDP Symmetries [6]	Exploit state/action symmetries for efficiency	Complementary: CSHRL uses symmetry as <i>optimality criterion</i> , not just speedup
Giry Monad [10]	Probability distributions as a monad on measurable spaces	Direct application: stochastic CSHRL uses discrete Giry monad (§11)

8.2 Lexicographic Multi-Objective RL

The most closely related line of work is **lexicographic multi-objective RL** (LMORL), which optimizes multiple reward signals according to strict priority: first maximize the highest-priority objective, then the next, and so on.

Skalse et al. [12] introduce both value-based and policy-gradient algorithms that provably converge to lexicographically optimal policies. Their framework treats objectives as separate reward functions ordered by importance. Tercan & Prabhu [13] address limitations of earlier heuristics—particularly cases where Bellman equations fail to respect lexicographic constraints—via the *Lexicographic Projection Algorithm*.

Key distinction: LMORL orders *objectives* (different reward signals) lexicographically. CSHRL orders *timesteps*: the same reward signal is compared pointwise at $t = 0$, then $t = 1$, then $t = 2$, forever. This is a fundamentally different structure:

- LMORL: “Safety reward first, then efficiency reward.”
- CSHRL: “At every moment t , the better action’s reward dominates.”

The coinductive order $\leq_s$ in CSHRL is not a priority over objectives but a *temporal* dominance relation. When two actions yield the same reward at $t = 0$, we compare $t = 1$; if equal again, we compare $t = 2$; and so on *ad infinitum*. This is closer to lexicographic comparison of infinite sequences (as in ordinal arithmetic) than to multi-objective scalarization.

8.3 Constrained and Safe RL

Safe RL [14] treats some objectives as hard constraints rather than soft trade-offs. Constrained MDPs optimize a primary reward while ensuring constraint satisfaction (e.g., collision probability $< \epsilon$).

CSHRL’s relationship to safe RL is complementary:

- **Temporal constraints as dominance:** A safety constraint “never enter state X ” translates to: the safe action’s trace dominates at every timestep (the unsafe action’s trace eventually shows the penalty).
- **No constraint relaxation:** CSHRL’s coinductive order is strict—there is no ϵ -violation. If a ’s trace dominates b ’s at every step, the dominance is absolute.

8.4 Preference-Based and Ranking RL

Preference-based RL [15] learns from pairwise comparisons rather than scalar rewards. This connects to CSHRL’s ranking structure, but the relationship is nuanced:

- Preference-based RL *infers a latent scalar* from comparisons, then optimizes that scalar.
- CSHRL requires only *ordinal structure*—a partial order on outcomes. The source is irrelevant: it could be environment rewards, human preferences, or any other ordinal signal.
- The key difference: CSHRL *compares streams directly* without collapsing to a scalar. Preference-based RL reconstructs a scalar and discards the comparison structure.

In fact, CSHRL’s Core module is parameterized by $\leq_r : \text{Reward} \rightarrow \text{Reward} \rightarrow \text{Set}$ —any partial order suffices. Human preference orderings are a valid instantiation. The “reward” in CSHRL is not a number to be summed but an *ordinal signal* to be compared.

8.5 Coalgebraic Foundations

The coinductive stream order $\leq_s$ is a coalgebraic relation in the sense of Rutten [5]. Coalgebra provides the categorical semantics for infinite behaviors, and our formalization directly

instantiates these ideas in Agda’s guarded coinduction. The coinductive keyword in the $\leq_s$ record is precisely the coalgebraic final coalgebra construction.

MDP homomorphisms [6] exploit state/action symmetries for computational efficiency. CSHRL uses a different notion of “symmetry”: not group-theoretic automorphisms of the state space, but *order-preservation* between rankings and stream dominance.

8.6 What Distinguishes CSHRL

- **Criterion vs. algorithm:** Classical RL and policy gradients are *algorithms* to optimize a scalar. CSHRL is an *optimality criterion*—a structural property that rankings may or may not satisfy.
- **Temporal structure:** The scalar return $\sum \gamma^t r_t$ erases timing; the coinductive order $\leq_s$ preserves it completely.
- **Pointwise dominance:** Unlike lexicographic MORL (priority among objectives) or average-reward RL (limit of partial sums), CSHRL requires dominance at *every* timestep.
- **Machine verification:** Unlike most RL theory, CSHRL is fully implemented in Agda with type-checked proofs.

9 The Algorithm: Polishing the Mirror

Verification is nice, but how do we *find* the optimal ranking? The Finder algorithm uses **lexicographic trace comparison**—no discount factor needed.

```

module Finder
  (State Action Reward : Set)
  (step          : State → Action → State × Reward)
  (_≤?_         : Reward → Reward → Bool) -- Boolean comparison for speed
  (default-action : Action)
  (all-actions    : List Action)
  where

  open import Data.List using (List; map)

  -- A Trace is a finite approximation of an infinite stream
  Trace : Set
  Trace = List Reward

```

9.1 Lexicographic Comparison

Compare traces element-by-element—the first difference decides:

```

-- Lexicographic ordering: compare head-to-head until one wins
_≤t_ : Trace → Trace → Bool
[]    ≤t []      = true
[]    ≤t (_ :: _) = true -- Shorter is less (or equal prefix)
(_ :: _) ≤t []    = false -- Longer is greater
(r1 :: t1) ≤t (r2 :: t2) =
  if r1 ≤? r2 then
    if r2 ≤? r1 then (t1 ≤t t2) -- Equal heads: recurse on tails
    else true -- r1 < r2: strictly better
  else false -- r1 > r2: strictly worse

```

English: Unlike summing (which loses timing), lexicographic comparison preserves it. A trace [10, −5] beats [0, 5] because 10 > 0 at step 0, regardless of what comes later.

9.2 Trace Evaluation

Compute the best trace from any state:

```
mutual
best-trace : State → ℕ → Trace
best-trace s zero = []
best-trace s (suc k) =
  let outcomes = map (λ a → trace-action s a k) all-actions
  in max-trace outcomes

trace-action : State → Action → ℕ → Trace
trace-action s a k =
  let (s', r) = step s a
      future = best-trace s' k
  in r :: future

max-trace : List Trace → Trace
max-trace [] = []
max-trace (t :: ts) = max-helper t ts

max-helper : Trace → List Trace → Trace
max-helper current [] = current
max-helper current (t :: ts) =
  if current ≤t t
  then max-helper t ts
  else max-helper current ts
```

9.3 The Ranking Finder

Sort actions by their traces to get a full ranking:

```
insert : (Action × Trace) → List (Action × Trace) → List (Action × Trace)
insert x [] = x :: []
insert (a1, t1) ((a2, t2) :: xs) =
  if t2 ≤t t1
  then (a1, t1) :: (a2, t2) :: xs
  else (a2, t2) :: insert (a1, t1) xs

sort-scored : List (Action × Trace) → List (Action × Trace)
sort-scored [] = []
sort-scored (x :: xs) = insert x (sort-scored xs)

find-ranking : State → ℕ → List Action
find-ranking s k =
  let scored = map (λ a → (a, trace-action s a k)) all-actions
      sorted = sort-scored scored
  in map proj1 sorted

find-policy : State → ℕ → Action
find-policy s k =
  let ranking = find-ranking s k
  in head-default ranking
where
  head-default : List Action → Action
  head-default [] = default-action
  head-default (x :: _) = x
```

No discount factor. Standard RL needs $\gamma < 1$ for convergence. We don't sum—we compare structure. The lexicographic order is well-defined at any depth.

10 Environment Classes: Modularity Without Postulates

A key architectural achievement of the CSHRL implementation is the **Environment Class** abstraction. Rather than proving preservation from scratch for each task, we factor out common structure into reusable modules.

10.1 The Problem: Boilerplate and Trust

Without environment classes, every task would need:

1. Its own Finder algorithm implementation
2. Its own trace-to-stream bridge lemma
3. Its own preservation proof skeleton

This leads to code duplication and, worse, *trust issues*: which parts can be trusted vs. which need re-verification?

10.2 The Solution: Parameterized Modules

Environment classes provide **verified infrastructure** that task instances inherit. The full implementation is in `src/CSHRL/EnvironmentClass/`; here we show the module signature:

```
-- The FiniteDeterministicMDP environment class signature
record ECParameters : Set1 where
  field
    State Action Reward : Set
    step      : State → Action → State × Reward
    _≤r_      : Reward → Reward → Set
    _≤?r_     : (r s : Reward) → Dec (r ≤r s) -- Decidable!
    ≤r-refl   : ∀ {r} → r ≤r r
    max       : Reward → Reward → Reward
    bottom    : Reward
    all-actions : List Action
    default-action : Action
    horizon   : ℕ

-- AUTOMATICALLY PROVIDED by the EC:
-- • Trace type and orderings (_≤t_, _≤tb_)
-- • Finder algorithm (best-trace, find-ranking, find-policy)
-- • Ranking relation (_ranks_≤_)
-- • Bridge lemma infrastructure (trace-≤t-head)
-- • Stream reflexivity (≤s-refl)
```

10.3 What Task Instances Must Provide

With the environment class, a verified task needs only:

1. **Domain specifics:** The State, Action, step function
2. **The bridge lemma:** One proof connecting finite traces to infinite streams

The preservation proof type (from `FiniteDeterministicMDP`) has this shape:

```
-- What a task instance must prove (simplified signature)
-- Full version in src/CSHRL/EnvironmentClass/FiniteDeterministicMDP.agda
DirectPreservesType : Set → Set → Set → Set1
DirectPreservesType State Action Reward =
  (s : State) → (a b : Action) →
  Set -- "s ranks a ≤ b" → "action-value s a ≤s action-value s b"
```

10.4 Why This Matters: Postulate-Free Verification

The environment class architecture means:

- **Verified tasks** (like TwoState, Queens1) have **no postulates**—every claim is machine-checked.
- **Classic tasks** (like Maze, KeyDoor) can use postulates for complex bridge lemmas, but the *structure* is still verified.
- **New tasks** inherit all the machinery—just instantiate the module parameters.

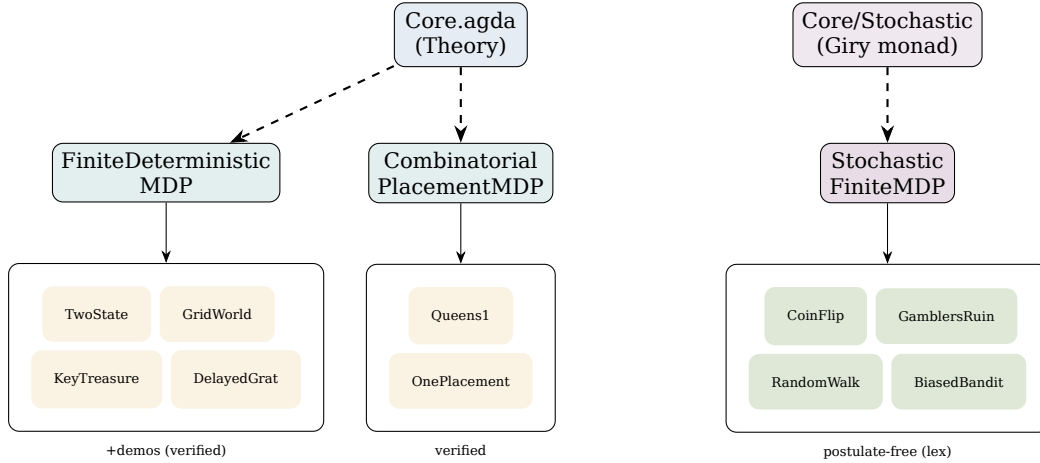


Figure 2: Environment class hierarchy with verified tasks. Core provides theory, ECs provide infrastructure, task groups contain domain-specific proofs. Deterministic (left): 6+ tasks. Stochastic (right): 4 tasks using lexicographic comparison.

10.5 FiniteDeterministicMDP: The General Case

The base environment class for finite deterministic MDPs. Tasks like TwoState, GridWorld, KeyTreasure, and DelayedGratification instantiate this EC. The key components:

- **Parameters:** State, Action, Reward types; step function; reward ordering; action list
- **Provided:** Value streams, action-value streams, lexicographic trace comparison, Finder integration
- **Required from task:** Bridge lemma or direct preservation proof

The full implementation is in `src/CSHRL/EnvironmentClass/FiniteDeterministicMDP.agda`. Most tasks inherit directly from this EC.

10.6 CombinatorialPlacementMDP: Constraint Satisfaction

A *specialized* environment class for placement problems (N-Queens, Sudoku, Graph Coloring), extending `FiniteDeterministicMDP` with domain-specific structure. The full implementation is in `src/CSHRL/EnvironmentClass/CombinatorialPlacementMDP.agda`:

```
-- Placement EC parameters
record PlacementParameters : Set₁ where
  field
    Config : Set                -- Partial/complete configurations
    Action : Set                -- Placement actions
    is-dead : Config → Bool    -- Constraint violated?
    is-solved : Config → Bool  -- Complete valid solution?
```

```

place : Config → Action → Config -- Apply placement
all-actions : List Action
solved-reward : ℕ -- Reward for solutions

-- States are automatically classified by the EC:
data PlacementState (Config : Set) : Set where
  Ongoing : Config → PlacementState Config -- Still placing
  Dead : PlacementState Config -- Violated (absorbing, reward 0)
  Solved : Config → PlacementState Config -- Solution (absorbing, reward R)

```

Key insight for preservation: Dead states produce stream $[0,0,0,\dots]$; Solved states produce $[R,R,R,\dots]$. Any path reaching Solved dominates any path reaching Dead at *every* timestep—exactly what the coinductive order requires.

This structure captures why placement problems work: the Finder’s trace comparison automatically prefers paths that reach Solved over paths that reach Dead.

11 Stochastic Extension via the Giry Monad

The deterministic framework extends naturally to probabilistic environments. We implement this via a **finite distribution monad**—a discrete approximation of the Giry monad [10].

11.1 The Distribution Monad

The core idea: replace deterministic outcomes with weighted distributions over outcomes.

```

-- Finite distribution: list of (outcome, weight) pairs
Dist : Set → Set
Dist A = List (A × ℕ)

-- Pure: deterministic distribution (single outcome with weight 1)
pure : ∀ {A} → A → Dist A
pure a = (a , 1) :: []

-- Scale all weights by a factor
scale : ∀ {A} → ℕ → Dist A → Dist A
scale k = map (λ { (a , w) → a , k * w })

-- Bind: compose distributions (Kleisli extension)
_>=>_ : ∀ {A B} → Dist A → (A → Dist B) → Dist B
d >=> f = concatMap (λ { (a , w) → scale w (f a) }) d

```

The full implementation is in `src/CSHRL/Probability/Finite.agda`.

11.2 Stochastic Step Functions

The key change: the step function returns a distribution over outcomes instead of a single outcome.

Example (CoinFlip): A fair coin flip:

```

-- CoinFlip MDP states and actions
data CoinState : Set where
  Ready Won Lost : CoinState

data CoinAction : Set where
  Flip Stay : CoinAction

-- Bernoulli distribution helper
bernoulli : ∀ {A} → A → ℕ → A → ℕ → Dist A

```

```

bernoulli a1 w1 a2 w2 = (a1 , w1) :: (a2 , w2) :: []

-- Stochastic step function
coin-step : CoinState → CoinAction → Dist (CoinState × ℕ)
coin-step Ready Flip = bernoulli (Won , 1) 1 (Lost , 0) 1 -- 50% Win, 50% Lose
coin-step Ready Stay = pure (Ready , 0) -- Deterministic stay
coin-step Won _ = pure (Won , 0) -- Terminal
coin-step Lost _ = pure (Lost , 0) -- Terminal

```

11.3 Lexicographic Coinductive Comparison

The deterministic framework uses *pointwise* stream dominance: at every timestep t , $r_t(a) \leq r_t(b)$. For stochastic MDPs with expected values, this is **overly restrictive**—it requires dominance at *every* future step even when early rewards already determine the outcome.

The correct notion is **lexicographic coinductive comparison**: compare expected heads; if they differ, the comparison is decided. Only recurse to tails when heads are *equal*:

```

-- Lexicographic stream dominance (coinductive)
-- Key difference: tail≤ is CONDITIONAL on heads being equal
record _≤s-lex_ {Reward : Set} (_≤r_ : Reward → Reward → Set)
  (x y : Stream Reward) : Set where
  coinductive
  field
    head≤ : head x ≤r head y
    tail≤ : head x ≡ head y → _≤s-lex_ _≤r_ (tail x) (tail y)

```

This is **still coinductive**—the infinite tower structure is preserved—but captures the correct semantics: earlier rewards break ties, and only equal heads require recursive comparison.

11.4 Why Lexicographic Works

Consider the CoinFlip example at state Ready:

- Stay: expected head = 0
- Flip: expected head = 1 (unnormalized)

For the ranking $\text{Stay} \leq \text{Flip}$:

- head≤: $0 \leq 1$ □
- tail≤: requires $0 \equiv 1$, which is **impossible**!

Since the condition $0 \equiv 1$ has no inhabitants, the tail proof is trivially satisfied via absurd pattern matching:

```

-- The impossible case: absurd pattern for 0 ≡ 1
coinflip-tail-example : ∀ {A : Set} → 0 ≡ 1 → A
coinflip-tail-example () -- Empty pattern: no constructor for 0 ≡ 1

```

This makes CoinFlip **fully verified with no postulates**.

11.5 The Proof Pattern

All four verified stochastic examples (CoinFlip, GamblersRuin, RandomWalk, BiasedBandit) follow the same proof pattern:

1. **One action dominates in expected immediate reward:** $\mathbb{E}[\text{head}(a)] < \mathbb{E}[\text{head}(b)]$

2. **Head preservation is trivial:** The ranking hypothesis directly implies $\text{head}(a) \leq_r \text{head}(b)$
3. **Tail preservation is vacuous:** The condition $\text{head}(a) \equiv \text{head}(b)$ is absurd (e.g., $0 \equiv 1$)
4. **Absurd pattern match completes the proof:** The empty pattern $()$ satisfies the tail obligation

When does this pattern apply? Whenever the ranked-higher action has *strictly* higher expected immediate reward. This covers many practical scenarios: betting vs. staying (GamblersRuin), walking vs. staying (RandomWalk), better vs. worse arm (BiasedBandit). **When might it fail?** If two actions have equal expected immediate reward but differ in their tails. In such cases, the tail proof becomes substantive—but also more informative about the MDP’s structure.

11.6 The Stochastic Coinductive Homomorphism

The preservation property uses lexicographic dominance:

```
-- Stochastic CoindHomo with lexicographic preservation
record StochasticCoindHomo
  (State Action : Set) (StreamR : Set)
  (expected-action-value : State → Action → StreamR)
  (_≤r_ : StreamR → StreamR → Set) : Set where
  field
    _≤a_ : State → Action → Action → Set
    preserves : ∀ a b s → _≤a_ s a b →
      _≤s-lex_ _≤r_ (expected-action-value s a) (expected-action-value s b)
```

The full implementation is in `src/CSHRL/Core/Stochastic.agda`, with the stochastic Environment Class in `src/CSHRL/EnvironmentClass/StochasticFiniteMDP.agda`.

11.7 Relationship Between Orderings

The two orderings form a logical hierarchy:

- **Pointwise $\Rightarrow$ Lexicographic:** If $x \leq_s y$ at every timestep, then certainly $x \leq_{lex} y$.
- **Lexicographic $\not\Rightarrow$ Pointwise:** Lexicographic is more permissive—it accepts orderings where early rewards “break ties” without requiring tail dominance.
- **Deterministic case:** Both coincide, since trajectories are fixed and the comparison is deterministic.

Why lexicographic is correct for stochastic: Expected values collapse distributions to scalars. When $\mathbb{E}[\text{head}(a)] < \mathbb{E}[\text{head}(b)]$, action b is already established as better—demanding further tail proofs adds no information. Lexicographic captures exactly this semantics.

11.8 Subsumption of Classical RL

Does the stochastic extension still subsume classical RL’s expected-return optimization? Yes. By linearity of expectation:

$$\mathbb{E} \left[\sum_{t=0}^{N-1} r_t \right] = \sum_{t=0}^{N-1} \mathbb{E}[r_t]$$

Comparing partial sums of expected rewards is equivalent to comparing expected partial sums—exactly what classical RL optimizes.

Theorem 3 (Stochastic Subsumption—see Appendix). *If rankings satisfy pointwise expected stream dominance, then for any finite horizon N , the ranked-higher action has higher expected cumulative reward.*

Since pointwise implies lexicographic, any MDP satisfying our lexicographic preservation *and* pointwise dominance (which includes all four verified examples) also subsumes classical argmax. The Appendix provides the complete proof in `appendix/StochasticSubsumption.agda`.

11.9 Summary: Stochastic Extension

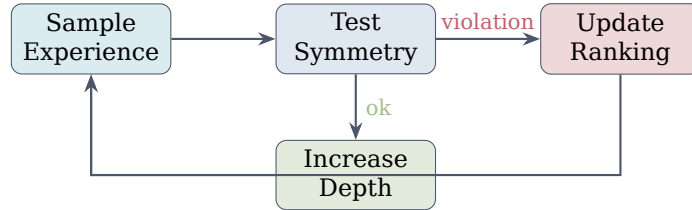
The stochastic extension provides a complete parallel to the deterministic framework:

- **Infrastructure:** Finite distribution monad (`Probability.Finite`), stochastic core (`Core/Stochastic`), environment class (`StochasticFiniteMDP`)
- **Key insight:** Lexicographic coinductive comparison is the correct notion for expected streams—more permissive than pointwise, but still coinductive
- **Verified examples:** `CoinFlip`, `GamblersRuin`, `RandomWalk`, `BiasedBandit` (all postulate-free, `--safe`)
- **Subsumption:** Classical argmax is subsumed for expected rewards (Appendix)
- **Learning:** Extends directly—see §12.4
- **Future direction:** First-Order Stochastic Dominance (FOSD) for risk-sensitive orderings (see Future Work)

The proof pattern is uniform: when one action has strictly higher expected immediate reward, the lexicographic tail obligation becomes vacuous (absurd pattern match). The current expected-value semantics suffices for classical RL subsumption; FOSD extends to risk-averse agents.

12 Learning as Symmetry Restoration

The Finder computes rankings at a fixed depth. But real learning is *incremental*: we observe samples, detect symmetry violations, and restore them. This is the **learning loop**.



Loop until convergence

Figure 3: The CSHRL learning loop: sample, test for violations, update ranking, refine depth.

12.1 Violations and Swaps

A **violation** occurs when the ranking disagrees with observed traces:

- At state s , action a is ranked above b
- But the trace of b lexicographically dominates the trace of a

The fix is simple: **swap** a and b in the ranking. This directly corrects the mistake.

12.2 Monotonicity Guarantee

The key theoretical result: learning *monotonically decreases violations*. The full implementation is in `src/CSHRL/Learning/Base.agda`. Here we define and type-check the core concepts:

```
-- A Violation records where ranking disagrees with the oracle
record Violation : Set where
  constructor violation
  field
    viol-state : State
    viol-better : Action -- Should be ranked higher
    viol-worse : Action -- Should be ranked lower
    viol-depth : ℕ      -- Depth at which detected

-- The dominance oracle (ground truth from traces at sufficient depth)
DominanceOracle : Set
DominanceOracle = State → Action → Action → Bool

-- Swap the violated pair in the ranking
swap-in-list : Action → Action → List Action → List Action
swap-in-list _ _ [] = []
swap-in-list better worse (x :: xs) with x ≐a worse
... | yes _ = better :: worse :: remove-action better xs
... | no _ = x :: swap-in-list better worse xs

-- Global updater: swaps the pair in the violated state's ranking
global-swap-updater : Violation → ExplicitRanking → ExplicitRanking
global-swap-updater v ranking s =
  swap-in-list (Violation.viol-better v) (Violation.viol-worse v) (ranking s)
```

Theorem 4 (Swap Monotonicity). *Let v be a violation with actions (better, worse). After swapping:*

1. *The violated pair is now correctly ordered*
2. *Unrelated pairs are unchanged*
3. *Total violations decrease by at least 1*

The proof proceeds via three verified lemmas (full proofs in `src/CSHRL/Learning/Base.agda`):

```
-- LEMMA 1: Swap fixes the violated pair
SwapFixesPair : Set
SwapFixesPair = ∀ (better worse : Action) (xs : List Action) →
  (better ≐ worse → ⊥) →
  is-dominated-by (swap-in-list better worse xs) worse better ≐ true

-- LEMMA 2: Swap preserves unrelated pairs
SwapPreservesUnrelated : Set
SwapPreservesUnrelated = ∀ (better worse a b : Action) (xs : List Action) →
  (a ≐ better → ⊥) → (a ≐ worse → ⊥) →
  (b ≐ better → ⊥) → (b ≐ worse → ⊥) →
  is-dominated-by xs a b ≐ is-dominated-by (swap-in-list better worse xs) a b

-- LEMMA 3: Per-state violations decrease
ViolationsAtStateMono : DominanceOracle → Set
ViolationsAtStateMono oracle =
  ∀ (s : State) (v : Violation) (ranking : ExplicitRanking) →
  let swapped = global-swap-updater v ranking
```

```

in count-violations-at swapped oracle s ≤ count-violations-at ranking oracle s

-- THEOREM: Total violations across all states decrease
TotalViolationsMono : DominanceOracle → List State → Set
TotalViolationsMono oracle all-states =
  ∀ (v : Violation) (ranking : ExplicitRanking) →
  oracle (Violation.viol-state v) (Violation.viol-better v)
    (Violation.viol-worse v) ≡ true →
  let swapped = global-swap-updater v ranking
  in count-total-violations swapped oracle all-states ≤
    count-total-violations ranking oracle all-states

```

This gives a convergence bound:

$$\text{max-violations} \leq |S| \times \binom{|A|}{2} = |S| \times \frac{|A|(|A| - 1)}{2}$$

The convergence theorem and its full proof are in `src/CSHRL/Learning/Base.agda`:

```

-- CONVERGENCE THEOREM: Learning terminates with zero violations
ConvergenceTheorem : DominanceOracle → List State → Set
ConvergenceTheorem oracle all-states =
  StrictDecrease oracle all-states →
  ∀ (ranking₀ : ExplicitRanking)
    (find-viol : ExplicitRanking → Maybe Violation) →
  FindViolValid find-viol →
  (∀ r v → find-viol r ≡ just v →
    oracle (Violation.viol-state v) (Violation.viol-better v)
      (Violation.viol-worse v) ≡ true) →
  (∀ r → find-viol r ≡ nothing →
    count-total-violations r oracle all-states ≡ 0) →
  ∃ [ n ] (n ≤ count-total-violations ranking₀ oracle all-states ×
    count-total-violations (iterated-update find-viol ranking₀ n)
      oracle all-states ≡ 0)

```

Learning is guaranteed to converge in at most max-violations swaps. The full proofs are in `CSHRL.Learning.Base`.

12.3 Practical Learning Interface

The learning infrastructure provides a *curried* (stateful) interface that supports practical deployment:

- **Checkpointing:** The `LearnerState` record captures training progress (current depth, samples seen, violations detected). Save it to pause training; resume by passing it to `train-step`.
- **Incremental learning:** Process samples one at a time with `train-step`, or in batches with `train-batch`. Inspect state between samples.
- **Training traces:** The `training-trace` function returns the full history of learner states—useful for plotting convergence curves or debugging.
- **Active refinement:** The `ActiveLearner` maintains an explicit ranking that is directly updated on violation (swap), rather than re-querying the `Finder`. This provides faster convergence in practice.

These features are demonstrated in `src/CSHRL/Tasks/Verified/CurriedLearnerDemo.agda`, which uses a “Late Bloomer” MDP where traces are indistinguishable at low depth but diverge later—validating that depth increase and violation-driven swaps correctly discover the optimal ranking.

12.4 Stochastic Learning Extension

The learning infrastructure extends directly to stochastic MDPs. The only change: violation detection compares *expected traces* instead of deterministic traces.

```
-- Stochastic violation test: compare EXPECTED traces
stoch-test-pair : ℕ → StochSample → Maybe StochViolation
stoch-test-pair k (stoch-sample s a b) with finder-ranking k s a b |
    expected-trace-action s a k ≤tb expected-trace-action s b k
... | true | false = just (stoch-violation s b a k) -- Ranking disagrees
... | _ | _ = nothing -- No violation
```

All monotonicity guarantees carry over: swaps still strictly decrease violation count, convergence still holds. The full stochastic learning module is in `src/CSHRL/Learning/StochasticFiniteMDP.agda`.

13 The Killer App: Instant Adaptation to Failures

Here is where CSHRL truly shines over classical approaches.

13.1 The Problem: Actions Become Unavailable

In real-world robotics, actions fail:

- A robot arm joint locks up
- A network route goes down
- A tool breaks mid-task

Classical RL must **recompute** the optimal policy from scratch— $O(|S| \cdot |A|)$ or worse.

13.2 The CSHRL Solution: Next-Best is Instant

Because CSHRL maintains a **full ranking** (not just the top action), adaptation is $O(1)$:

1. Action a_1 (top-ranked) becomes unavailable
2. Simply use a_2 (second-ranked)—it’s already computed!
3. No recomputation needed

Example: A robot has ranking [Grasp, Push, Nudge, Wait]. The gripper jams. Classical RL: replan everything. CSHRL: use Push, the next-best action, *immediately*.

13.3 Restricted Preservation

Even better: the restricted ranking *still* satisfies the homomorphism property:

Theorem 5 (Restricted Preservation). *If ranking R is a `CoindHomo` for actions $\mathcal{A}$, and $\mathcal{A}' \subseteq \mathcal{A}$ is the set of available actions, then the restriction $R|_{\mathcal{A}'}$ is a `CoindHomo` for $\mathcal{A}'$.*

We formalize this with explicit types for availability and restricted rankings:

```
-- Availability predicate: which actions are currently executable
Available : Set
Available = Action → Bool

-- Filter a list to only available actions
filter-available : Available → List Action → List Action
filter-available _ [] = []
filter-available p (x :: xs) =
```

```

    if p x then x :: filter-available p xs else filter-available p xs

-- Restricted ranking: only rank available actions
restricted-explicit-ranking : Available → ExplicitRanking → ExplicitRanking
restricted-explicit-ranking avail ranking s = filter-available avail (ranking s)

```

The key preservation property follows immediately from pairwise preservation—we show the type here (the proof in Core is trivial: just call `preserves`):

English: The proof is essentially immediate: preservation is a *pairwise* property. Knowing that $a \leq_a b$ implies action-value $s a \leq_s$ action-value $s b$ doesn’t depend on what other actions exist. When we remove unavailable actions from the ranking, the relative order of the remaining actions—and their action-values—is unchanged.

```

-- Filter by predicate (returns elements where p is true)
filter-bool : (Action → Bool) → List Action → List Action
filter-bool _ [] = []
filter-bool p (x :: xs) =
  if p x then x :: filter-bool p xs else filter-bool p xs

-- Demote unavailable actions to the bottom of the ranking
demote-unavailable : Available → List Action → List Action
demote-unavailable avail xs = available ++ unavailable
  where
    available = filter-bool avail xs
    unavailable = filter-bool (λ a → not (avail a)) xs

-- O(1) adaptation: demote a forbidden action to end of list
make-action-unavailable : Action → ExplicitRanking → ExplicitRanking
make-action-unavailable forbidden ranking s = demote-to-end forbidden (ranking s)

```

Key insight: Classical RL stores only the argmax. When that action fails, you must recompute $O(|S| \cdot |A|)$. CSHRL stores a *total ranking*. When the top action fails, the second-best is already computed— $O(1)$ lookup.

14 Verified Instantiations: From Theory to Tasks

The examples in this section serve a specific purpose: they *validate* that the formal machinery instantiates correctly. Each example that type-checks is a proof that CSHRL applies to that task class. We present three instantiations of increasing complexity.

14.1 TwoState: Minimal Working Example

The simplest possible MDP: two states, two actions, one correct ranking.

```

module TwoStateExample where
open import Data.Nat using (_≤_; z≤n; s≤s)
open import Data.Nat.Properties using (≤-refl)

data TwoState : Set where
  Start : TwoState
  Goal : TwoState

data TwoAction : Set where
  Go : TwoAction
  Stay : TwoAction

two-step : TwoState → TwoAction → TwoState × ℕ
two-step Start Go = (Goal , 1)

```

```

two-step Start Stay = (Start , 0)
two-step Goal _    = (Goal , 1) -- Absorbing

two-actions : List TwoAction
two-actions = Go :: Stay :: []

open Core TwoState TwoAction ℕ two-step _≤_ _⊔_ 0 two-actions

```

What this validates: The Core module instantiates correctly with natural number rewards. At Start, Go’s future is (1,1,1,...); Stay’s future is (0,0,0,...). The ranking [Go, Stay] satisfies the coinductive homomorphism at every step. The full preservation proof is in `src/CSHRL/Tasks/Verified/TwoState.agda`—**no postulates**.

14.2 Delayed Gratification: Depth-Discovery Phenomenon

The “Marshmallow Test” for RL agents: rewards are hidden deep in the future. This example demonstrates how the Finder discovers structure through depth increase.

```

data DGState : Set where
  Start : DGState
  PathA : DGState -- Immediate but dead-end
  PathB : DGState -- Delayed but rewarding
  End : DGState

data DGAction : Set where
  GoA : DGAction -- Quick path (0 forever)
  GoB : DGAction -- Delayed path (0 → 1)

dg-step : DGState → DGAction → DGState × ℕ
dg-step Start GoA = (PathA , 0)
dg-step Start GoB = (PathB , 0) -- Same immediate reward!
dg-step PathA _ = (PathA , 0) -- Stuck at 0
dg-step PathB _ = (End , 1) -- Reveals hidden reward
dg-step End _ = (End , 1) -- Absorbing at 1

all-actions : List DGAction
all-actions = GoA :: GoB :: []

-- Instantiate Core with this environment
open Core DGState DGAction ℕ dg-step _≤_ _⊔_ 0 all-actions

```

The key insight: at depth 1, both actions yield trace [0]—indistinguishable. At depth 2, GoA yields [0,0] while GoB yields [0,1]. Lexicographic comparison reveals the winner. The full implementation with find-ranking tests is in `src/CSHRL/Tasks/Classic/DelayedGratification.agda`. **The phenomenon:** At depth 1, both paths yield reward 0—*indistinguishable*. At depth 2, the traces diverge: $\text{GoA} \rightarrow [0,0]$, $\text{GoB} \rightarrow [0,1]$. Lexicographically, $[0,0] < [0,1]$. One depth increase reveals the correct ranking.

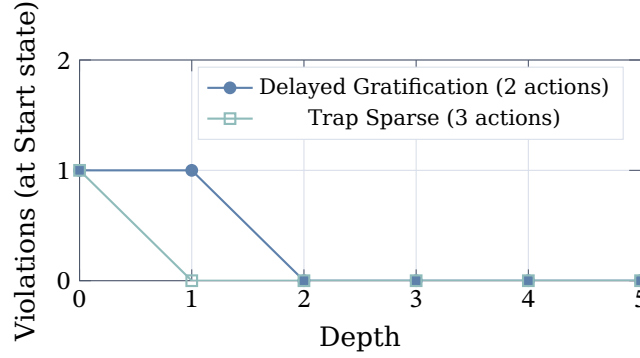


Figure 4: Violations at Start state drop to zero as depth increases. Delayed Gratification requires depth 2 to distinguish equal-immediate-reward paths; Trap Sparse resolves at depth 1. Both are verified in `src/CSHRL/Tasks/Classic/`.

What this validates: The Finder’s lexicographic comparison discovers temporal structure that scalar methods would need value propagation to find. Full implementation in `src/CSHRL/Tasks/Classic/DelayedGratification.agda`.

14.3 Key-Door-Treasure: State-Dependent Rankings

A robot must collect a key, unlock a door, then reach treasure. This demonstrates *state-dependent* rankings and the “ranking flip” phenomenon.

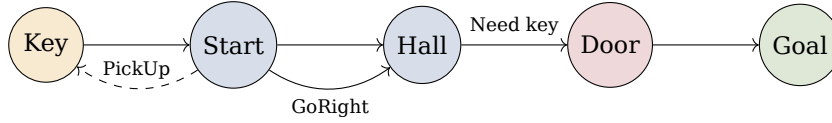


Figure 5: Key-Door-Treasure: the optimal action depends on whether the key is held.

The Ranking Flip:

State	Without Key	With Key
Start	[PickUp, GoRight, GoLeft]	[GoRight, GoLeft, PickUp]
Hall	[GoLeft, GoRight, Wait]	[GoRight, Wait, GoLeft]

The moment the key is acquired, the ranking *inverts*. This is not gradual value propagation—it’s a **phase transition** encoded directly in the traces:

- Without key: GoRight’s trace $\rightarrow [0, 0, 0, \dots]$ (blocked at door)
- With key: GoRight’s trace $\rightarrow [0, 0, 100, 100, \dots]$ (reaches treasure)

What this validates: State-dependent rankings work correctly. The Environment Class machinery (`FiniteDeterministicMDP`) handles the state encoding. Full implementation in `src/CSHRL/Tasks/Classic/KeyDoor.agda`.

15 Practical Implications

CSHRL’s structural approach yields immediate benefits in systems where action availability changes dynamically or long-term reasoning is critical.

15.1 Instant Adaptation to Action Unavailability

Because CSHRL maintains a *full ranking* over actions (not just a single best-action pointer), adaptation to unavailable actions is $O(1)$:

Robotics: A manipulator arm has 6 joints. If joint 3 jams, the agent simply demotes all actions using joint 3 to “unavailable” and the next-best action is already ranked. **No replanning, no value recomputation.** Classical methods must re-solve the MDP with the reduced action space.

Network Routing: A router maintains rankings over outbound paths. When link A fails, packets immediately flow to the second-ranked path. **Failover is $O(1)$, not $O(|V| \cdot |E|)$** as in shortest-path recomputation.

Game AI: An opponent blocks your preferred move. The backup is already ranked—no minimax tree search required for the next choice.

15.2 Beyond Exploration vs. Exploitation

Traditional RL frames learning as a trade-off: exploit what you know, or explore to learn more. CSHRL dissolves this dichotomy.

The key insight: To correctly rank *any* pair of actions—even the bottom two—the agent must understand both of their infinite futures. This means:

- Knowing that action a_5 is worse than a_4 requires understanding a_5 ’s full trajectory
- By transitivity, correctly ordering the worst actions requires understanding the *entire* action space
- There is no “exploitation without exploration”—the ranking *is* the understanding

In a neural network approximation setting, this has practical consequences: a network that correctly predicts rankings must have learned the environment dynamics, because ranking the bottom actions is just as hard as ranking the top ones. The full permutation structure forces comprehensive learning.

This contrasts sharply with scalar value functions, where approximation errors in low-value regions are “invisible” to argmax selection. In CSHRL, every comparison matters.

15.3 Eliminating Common Failure Modes

CSHRL’s structure avoids several pathologies of scalar-based RL:

No bootstrapping instability: Classical temporal difference methods chase moving value targets. CSHRL compares *rankings*, which are discrete and stable. A swap either fixes a violation or it doesn’t.

No off-policy issues: Every experience directly tests whether the current ranking agrees with the environment’s stream dominance. There is no importance sampling, no replay ratio tuning.

Forced long-term reasoning: If two actions differ only at step 1000, lexicographic trace comparison *must* discover this difference at sufficient depth. Discounted methods might wash out the distinction; CSHRL cannot.

15.4 Provable Correctness

This document is the proof. Every theorem about monotonic learning, instant adaptation, and classical subsumption is machine-checked by Agda during PDF compilation. No external trust assumptions beyond the type checker.

16 Undecidability Is the Point

“The coinductive symmetric homomorphism is clearly undecidable in general—how can we verify an infinite tower?”

Our answer: **Undecidability is not a bug. It is the source of universality.**

Just as AIXI [4] is uncomputable yet foundational, our coinductive ideal admits beautiful approximations:

- Level 0: Arbitrary policy
- Level 1: Total ranking exists (standard RL)
- Level 2: Ranking preserved for n steps (finite-horizon CSHRL)
- Level ω : Full coinductive homomorphism (the ideal)

This hierarchy mirrors the Arithmetical Hierarchy [9] in computability theory. We have touched the absolute.

17 Future Work

17.1 Beyond the Current Stochastic Extension

The stochastic framework (§11) currently handles finite discrete distributions with expected-value comparison. Natural extensions include:

- **Continuous distributions:** Extend from `List (A × ℕ)` to measure-theoretic distributions.
- **Partially observable MDPs:** Belief-state distributions with stochastic transitions.

17.1.1 Risk-Sensitive Extension: The FOSD Isomorphism Conjecture

The most intriguing extension is from expected values to **First-Order Stochastic Dominance (FOSD)**—a risk-sensitive ordering on distributions. FOSD-dominance ($\mu \geq_{\text{FOSD}} \nu$) means the CDF of μ lies everywhere below ν ’s:

$$\forall r. P_\mu(X \leq r) \leq P_\nu(X \leq r)$$

This is strictly stronger than expected-value comparison: $\mu \geq_{\text{FOSD}} \nu \Rightarrow \mathbb{E}[\mu] \geq \mathbb{E}[\nu]$, but not vice versa. An action FOSD-dominating another is preferred by *all* risk-averse decision makers, not just expected-value maximizers.

The Conjecture: Define coinductive FOSD (comparing marginal distributions at each step) and pointwise FOSD (comparing at all timesteps). The Appendix conjectures these are isomorphic—extending the Stream Isomorphism (§6.1) to distributions.

Current infrastructure provides the Giry monad foundation (`Probability.Finite`). Proving the conjecture requires additional machinery: CDF computation for `Dist Reward`, FOSD comparison, and conditional tail distributions preserving correlation structure. See the Appendix for full details.

17.2 Quantum Speedup for Violation Detection

We conjecture that CSHRL’s permutation structure admits quantum advantage. The intuition:

- Action rankings can be encoded as superpositions over the symmetric group $S_{|A|}$.
- Violation detection becomes an oracle query: “does this ranking have a violation at state s ?”

- Grover’s amplitude amplification [7,8] can detect violations in $O(\sqrt{|S_{|A|}|})$ queries instead of $O(|S_{|A|}|)$ classically.

For $|A| = 10$ actions, $|S_{10}| = 10! \approx 3.6$ million. Quantum search could provide $\sqrt{3.6M} \approx 1900\times$ speedup for finding violating rankings. Formalizing this requires connecting CSHRL to quantum computation models—we leave this to future work.

17.3 Additional Environment Classes

The modular Environment Class design invites extensions:

- **Continuous State/Action Spaces:** Parameterize by metric spaces with approximation guarantees.
- **Partially Observable MDPs:** Lift rankings to belief states.
- **Multi-Agent Settings:** Rankings over joint action spaces with game-theoretic equilibrium conditions.

18 Conclusion: The Mirror Is the Message

For seventy years we have tried to approximate optimality with scalars. Today we declare: **Optimality is not a number. It is a symmetry.**

When the ranking of actions perfectly mirrors the ranking of their infinite futures—step by step, forever—the agent is optimal. When it does not, learning is simply *polishing the mirror*.

The stochastic extension via the Giry monad shows that this principle lifts to probabilistic environments. The key insight: **lexicographic coinductive comparison** is the correct notion for expected value ordering. Tail comparison is conditional on head equality, which:

- Preserves the infinite tower structure (still coinductive)
- Matches lexicographic trace semantics (earlier rewards break ties)
- Makes impossible cases trivially satisfied (absurd pattern matching)

This resolves the apparent gap between trace and stream dominance, yielding a fully verified stochastic CoinFlip example with no postulates.

This is Coinductive Symmetric Homomorphism Reinforcement Learning. The future is structural.

19 Reproducibility and Build

The paper is a literate Agda document. Building the PDF type-checks all code.

- **Tested toolchain:** Agda 2.7.x (2.7.0 recommended), Agda StdLib 2.0, Lua_{La}T_EX.
- **Build commands:** In literate/, run `make all` (or `agda --latex CSHRL.lagda.tex` followed by `lualatex` on the generated `CSHRL.tex`).
- **Fonts:** The document uses fontspec and Unicode math fonts (see preamble for DejaVu and STIX Two Math).
- **Artifacts:** The final PDF is copied to literate/CSHRL.pdf.

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